

Customer Behavior Analysis

A deep dive into retail/e-commerce transaction data — uncovering purchase patterns, discount effectiveness, and customer segments to drive smarter business decisions.



994K

Total Revenue

5K

Total Customers

199

Avg. Order Value

3.6

Avg. Rating

About the Dataset

Dataset Profile

Retail / E-commerce structured tabular data at the **customer transaction level**.

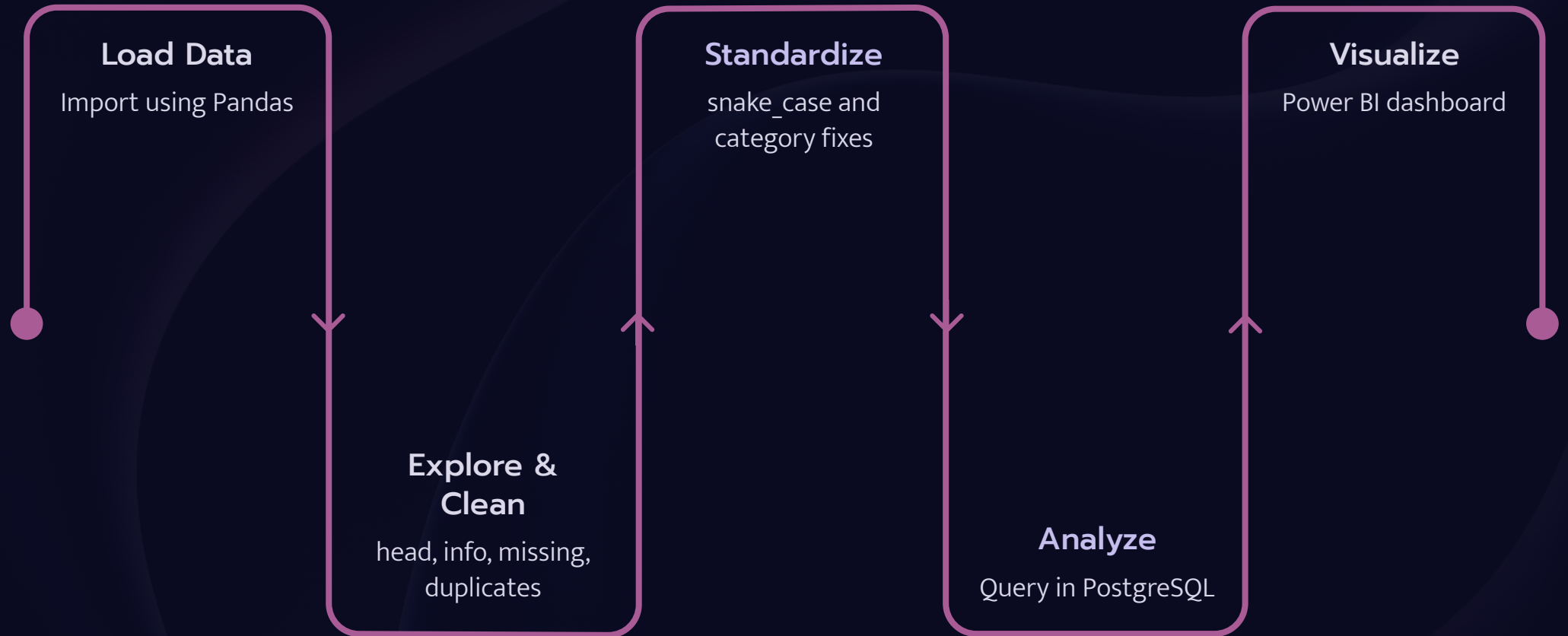
- **Numeric:** Age, Purchase Amount, Previous Purchases
- **Categorical:** Category, Payment Method, Season

Why It Matters

This data helps businesses:

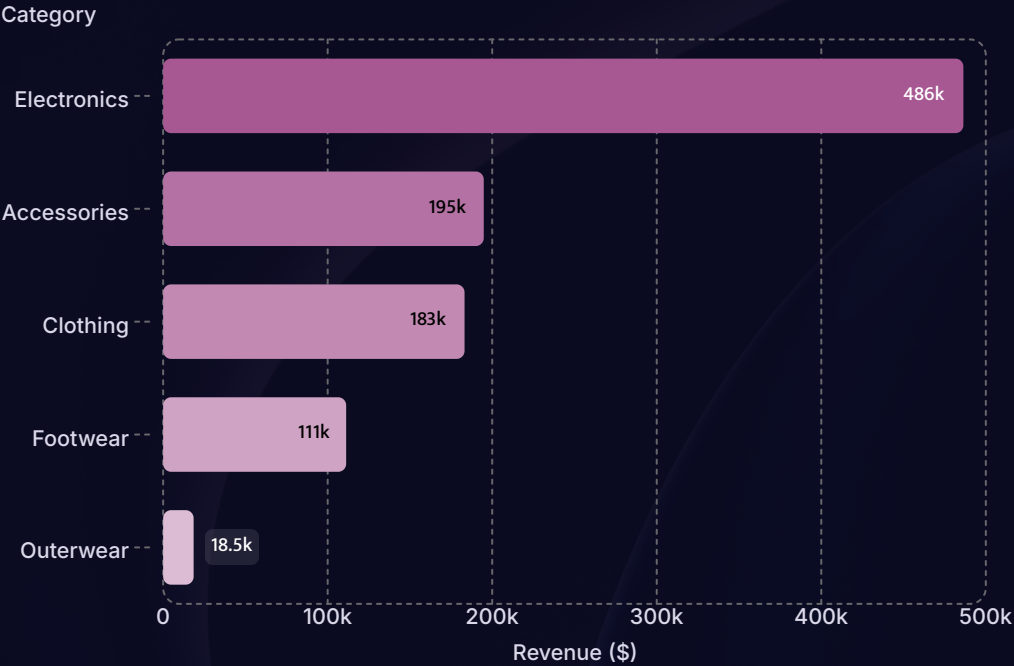
- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Improve marketing campaigns
- Increase customer lifetime value (CLV)

EDA & Data Pipeline



Python was used for exploratory data analysis and cleaning before exporting to PostgreSQL for structured querying and Power BI for visualization.

Revenue by Category & Gender



Key Findings

Electronics dominates with **\$486K** — nearly 2.5× the next category.

Revenue by Gender

Female \$352K — 39.3%	Male \$391K — 35.4%
Other \$251K — 25.3%	

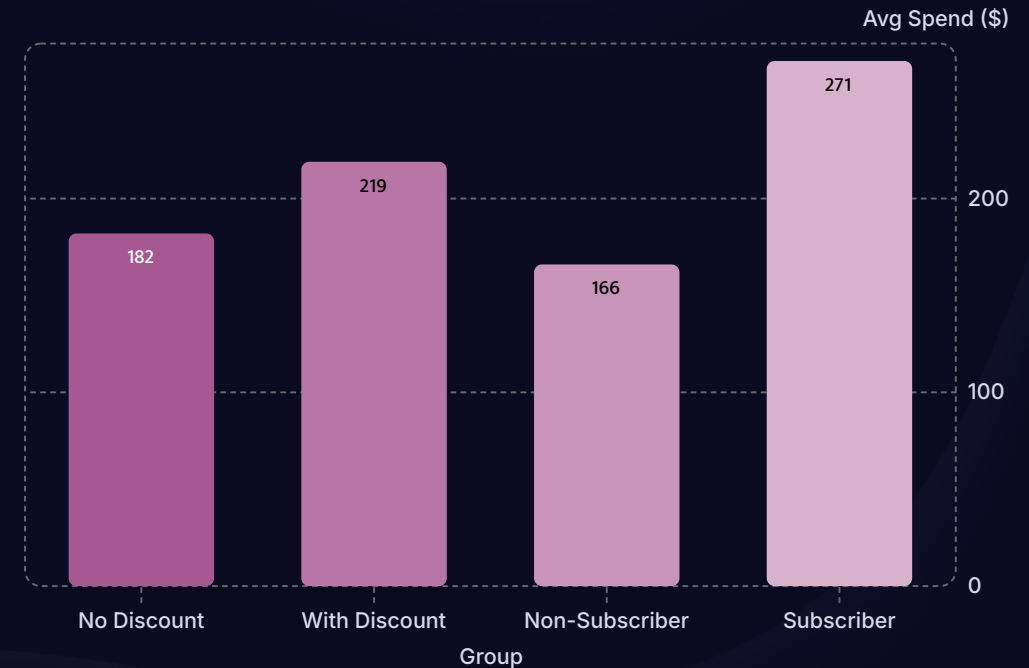
Discounts & Subscriptions

Do Discounts Increase Spend?

Yes — customers **with** discounts spend **\$219 avg** vs. \$182 without. Total revenue is nearly equal (\$492K vs. \$502K), suggesting discounts attract high-value buyers.

Subscribers vs. Non-Subscribers

Subscribed customers (1,579) spend **\$271 avg** vs. \$166 for non-subscribers (3,421) — a **63% higher average spend**.



Customer Segmentation

Loyal

3,537 customers

High previous purchase count

Returning

902 customers

Moderate repeat buyers

New

561 customers

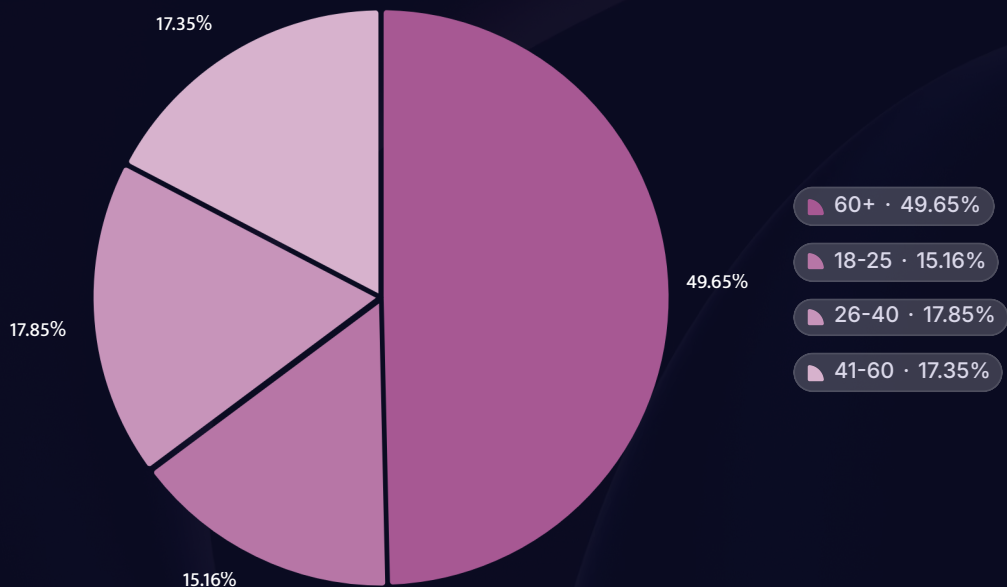
First-time or low-frequency

Of repeat buyers (5+ previous purchases), **1,190 subscribe** vs. 2,763 who don't — a significant upsell opportunity.

CUSTOMER LOYALTY PROGRAM



Revenue by Age Group & Season



Age Insight

The **60+ age group** contributes the most revenue at \$493K — nearly 50% of total.

Seasonal Revenue

Spring

\$243K

Winter

\$241K

Summer

\$237K

Autumn

\$213K

Product Ratings & Top Performers

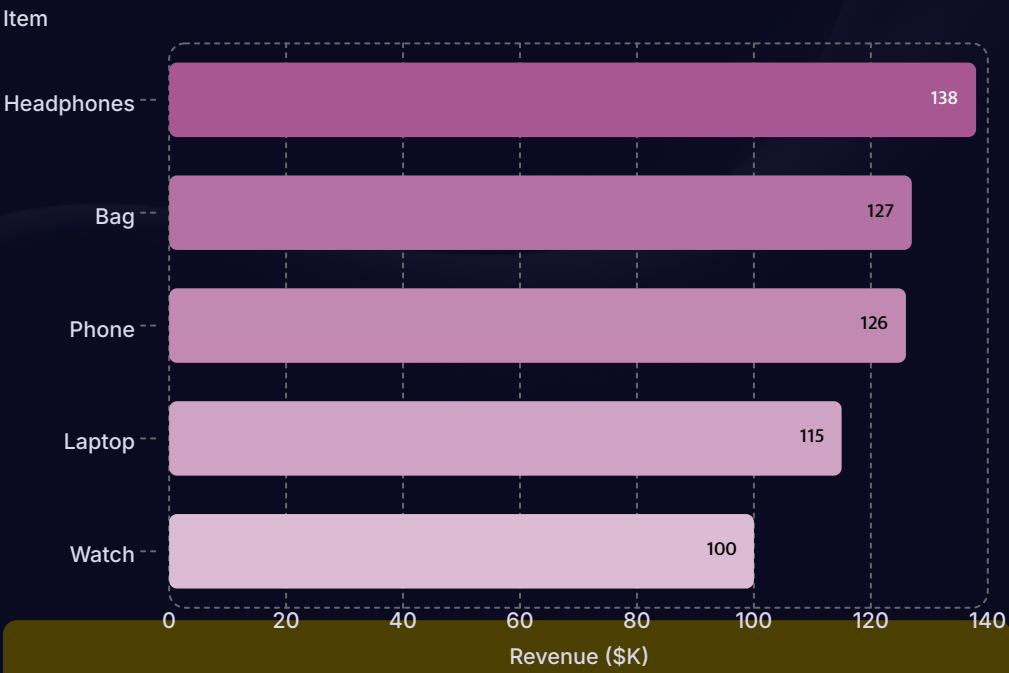
Top 5 by Rating

Item	Avg Rating
Gloves	3.9
Sandals	3.8
Sunglasses	3.8
Dress	3.8
Sweater	3.8

Bottom 5 by Rating

Item	Avg Rating
Phone	2.9
Headphones	3.1
Watch	3.1
Laptop	3.2
Bag	3.2

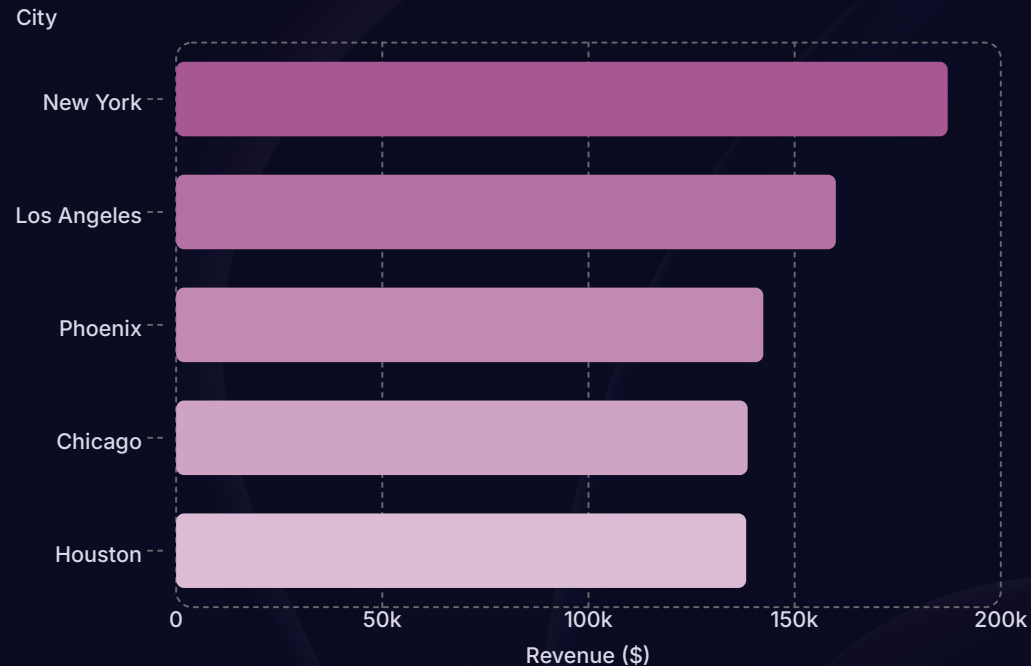
Top 5 by Revenue



⚠️ Electronics items (Phone, Headphones, Laptop) generate top revenue but have the **lowest ratings** — a quality gap to address.

Location, Payment & Shipping

Top 5 Locations by Revenue



Payment Methods

Cash, Credit Card, and Debit Card each drive **~\$225–253K** in revenue. Venmo and Bank Transfer are minimal.

Shipping: Express Wins

Express shipping averages **\$358** per order vs. **\$341** for Standard — higher-spend customers prefer faster delivery.



Power BI Dashboard

Interactive dashboards consolidate all findings — revenue by category, age group, gender, location, season, payment method, and shipping type — into a single decision-making view.

